

# DevOps Learners Assignment

**Topics Covered:** Git, GitHub, AWS EC2, Load Balancers, S3 Buckets & Static Websites, Shell Scripting (Nginx Setup), AWS AMIs & Volume Snapshots

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## Section 1: Git and GitHub (6 Questions)

### 1. What is the difference between Git and GitHub? Explain with examples.

Git is a version control system that works locally on our system by using command line interface, here we can perform tasks by using commands.

GitHub is a cloud based platform where Git repositories are stored, it stores the code and allows the teams to collaborate, review the code and manage projects remotely.

### 2. Write the Git commands for the following scenario:

- Initialize a new repository  
`git init`
- Add all files to the staging area  
`git add filename / git add .`
- Commit with a message "Initial commit"  
`git commit -m "Initial commit"`
- Push the code to a GitHub repository.  
`git remote add origin <repository-url>`  
`git push origin main`

### 3. How would you resolve a merge conflict in Git?

- Provide a simple situation where conflicts occur and how to solve them.

Merge conflict happens when two people or two branches try to make changes in the same part of a file, git doesn't know which change is correct and it asks to decide which version to be saved.

For example, me and prasanthi working on the same project file "index.html".

I have edited the file as

```
<h1>Hi, this is Mourya</h1>
```

Prasanthi edited the same file and same line as

```
<h1>Hi, this is Prasanthi</h1>
```

When we try to merge the changes, git gets confused as we try to change the same line and it will show the conflict inside the file. Now we need to decide to keep one version or we can combine both by editing the file manually.

```
<h1>Hi, this is Mourya and Prasanthi</h1>
```

Then we need to add and commit the file to complete the merge.

#### 4. Explain the purpose of **.gitignore** file with an example.

It is a special file where we list the unnecessary, large or sensitive files or folders that git should ignore.

For example, you are working on a project, and the project folder contains 'code.txt', 'notes.txt', 'passwords.txt', 'node\_modules/', and you want git to track 'code.txt', 'notes.txt' and ignore 'passwords.txt', 'node\_modules/'.

Create .gitignore file and add 'passwords.txt', 'node\_modules/' in it, so that git will completely ignore them, and it won't commit them, push them to github.

#### 5. Explain the differences among the following Git operations:

- **git pull**

It gets the latest changes (like new version of the project) from the git repository and merges it into the local files (project files).

- **git fetch**

It checks for the new changes and downloads them, but not automatically merges it into our local files. It lets us see what the new changes are made without disturbing our files.

- **git clone**

It downloads full copy of a remote project from GitHub to our local machine. Git clone was the initial step we need to do before starting to work on a project.

#### 6. What are the different types of Git reset (**--soft**, **--mixed**, **--hard**) and what does each one do?

Git reset is used to move the project backward to a previous point (to old commit), we can choose how much we want git to undo like just the commits or commits and staging area or commits, staging area and files too.

1. Git reset –soft: It only moves the commit pointer backwards, the code and files are not touched. The changes will be ready to commit again.
2. Git reset –mixed: Moves the commit pointer backward and unstages the files. The code and changes made will stay same in the working directory.
3. Git reset –hard: Moves the commit pointer backward, unstages the files and also delete changes from the working directory. Your project looks exactly like the old commit, clean but dangerous because we might loose the work.

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## Section 2: AWS EC2 and Load Balancers (4 Questions)

### 7. Describe the types of EC2 instances available (On-Demand, Reserved, Spot, Savings Plan) and explain the differences among them.

- 1) On-demand: Pay for only what we use (by hour or second)  
No long-term commitment.  
We can start and stop at any time.  
Good for testing & development, short-term projects.
- 2) Reserved: Commit to use the instance for 1 or 3 years.  
AWS gives a big discount (Cheaper than On-Demand).  
Good for applications that run 24/7, Long-term projects.
- 3) Spot Instances: Bid for un-used AWS Capacity at very low prices.  
AWS can stop it at anytime if someone else need it.  
Good for flexible, fault-tolerant jobs, not for critical systems.
- 4) Savings Plan: Commit to use certain amount of compute power for 1 or 3 years.  
Get flexibility to switch instance types and regions  
Get discount similar to Reserved instance.  
Good for long-term projects with some flexibility like instance type and region.

## 8. What is an Elastic Load Balancer (ELB)?

Elastic Load Balancer is a service that automatically distributes incoming traffic across multiple servers. It helps to keep the applications fast, available and reliable even if one server is busy or goes down.

- Name different types of load balancers AWS offers and when to use them.

1) Application Load Balancer: Best for web applications, Works at application layer (layer 7), Understands HTTP/HTTPS

Good for websites, apps needs to route traffic differently.

2) Network Load Balancers: Best for very fast network traffic, works at transport layer (Layer 4). Understands TCP, UDP, TLS Protocols.

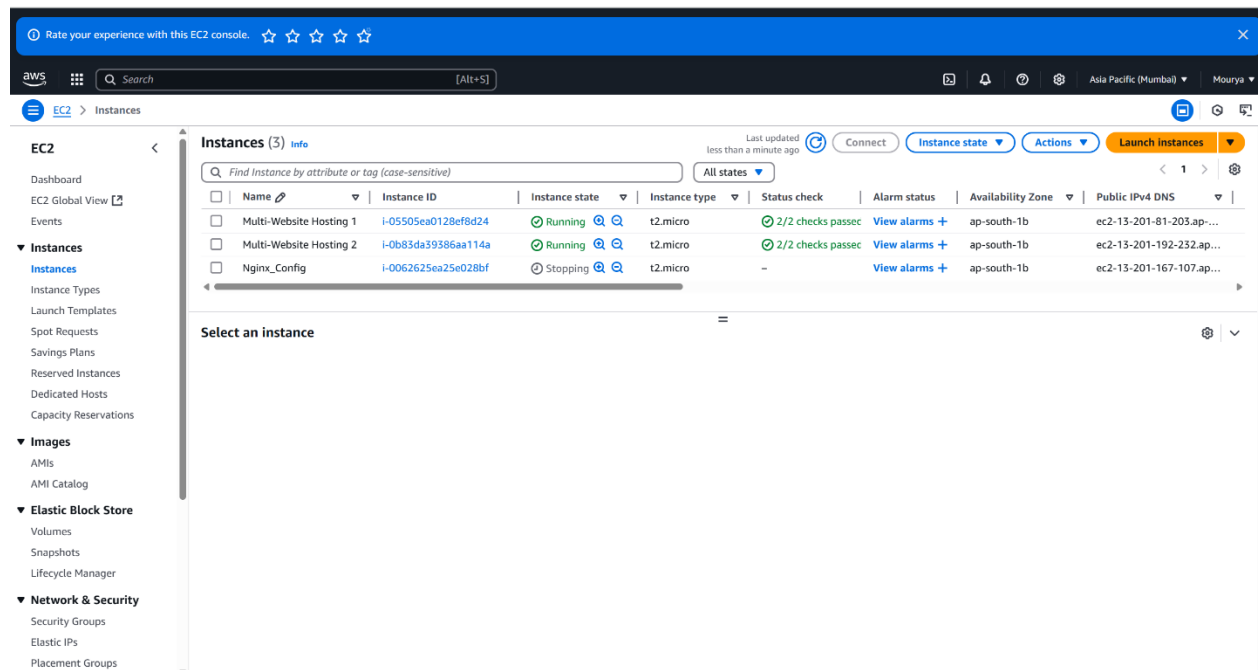
Good for gaming apps, real time trading apps, very high-speed networking apps.

3) Gateway Load Balancer: Works at Network layer (Layer 3), makes it easy to add third party security services to our network

Good for Large enterprise security setups, inserting firewalls or security checks without changing our app.

## 9. Explain how you would configure a load balancer to distribute traffic between two EC2 instances hosting the same website.

Step 1: Create two EC2 instances, make sure both the instances are in the same region. Allow http (port 80) traffic to both the instances & write your website html page on both the instances by using git bash.



The screenshot shows the AWS Management Console interface for EC2 instances. The left sidebar contains navigation links for EC2, Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, Lifecycle Manager, Network & Security, Security Groups, Elastic IPs, and Placement Groups. The main content area displays a table of instances with columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4 DNS. Three instances are listed: Multi-Website Hosting 1 (Running, t2.micro, 2/2 checks passed), Multi-Website Hosting 2 (Running, t2.micro, 2/2 checks passed), and Nginx\_Config (Stopping, t2.micro, 0/2 checks failed). Below the table, there is a 'Select an instance' dropdown menu.

| Name                    | Instance ID         | Instance state | Instance type | Status check      | Alarm status  | Availability Zone | Public IPv4 DNS          |
|-------------------------|---------------------|----------------|---------------|-------------------|---------------|-------------------|--------------------------|
| Multi-Website Hosting 1 | i-05505ea0128ef8d24 | Running        | t2.micro      | 2/2 checks passed | View alarms + | ap-south-1b       | ec2-13-201-81-203.ap...  |
| Multi-Website Hosting 2 | i-0b83da39386aa114a | Running        | t2.micro      | 2/2 checks passed | View alarms + | ap-south-1b       | ec2-13-201-192-232.ap... |
| Nginx_Config            | i-0062625ea25e028bf | Stopping       | t2.micro      | 0/2 checks failed | View alarms + | ap-south-1b       | ec2-13-201-167-107.ap... |

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AWS

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⚙️

Asia Pacific (Mumbai)

Mourya

EC2

Instances

i-05505ea0128ef8d24

EC2

Dashboard

EC2 Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

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Capacity Reservations

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Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Instance summary for i-05505ea0128ef8d24 (Multi-Website Hosting 1)

Info

🔄 Connect

Instance state

Actions

Updated less than a minute ago

Instance ID

i-05505ea0128ef8d24

IPv6 address

–

Hostname type

IP name: ip-172-31-0-73.ap-south-1.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

13.201.81.203 [Public IP]

IAM Role

–

IMDSv2

Required

Operator

–

Public IPv4 address

13.201.81.203 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-0-73.ap-south-1.compute.internal

Instance type

t2.micro

VPC ID

vpc-03c709098162d5641

Subnet ID

subnet-0cd1ec5e21897c12d

Instance ARN

arn:aws:ec2:ap-south-1:182563999454:instance/i-05505ea0128ef8d24

Private IPv4 addresses

172.31.0.73

Public IPv4 DNS

ec2-13-201-81-203.ap-south-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses

–

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name

–

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM Role

–

Owner ID

182563999454

Launch time

Sun Apr 27 2025 11:57:06 GMT+0530 (India Standard Time)

Rate your experience with this EC2 console.

☆ ☆ ☆ ☆ ☆

✕

AWS

Search

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Asia Pacific (Mumbai)

Mourya

EC2

Instances

i-05505ea0128ef8d24

EC2

Dashboard

EC2 Global View

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Instance Types

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Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

IMDSv2

Required

Instance ARN

arn:aws:ec2:ap-south-1:182563999454:instance/i-05505ea0128ef8d24

Managed

false

Operator

–

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM Role

–

Owner ID

182563999454

Launch time

Sun Apr 27 2025 11:57:06 GMT+0530 (India Standard Time)

Security groups

[sg-0d0e6e6d6918fd6d4 \(default\)](#)

▼ Inbound rules

Filter rules

< 1 >

| Name | Security group rule ID | Port range | Protocol | Source            | Security groups         |
|------|------------------------|------------|----------|-------------------|-------------------------|
| –    | sg-0818b1748213e8756   | 80         | TCP      | 0.0.0.0/0         | <a href="#">default</a> |
| –    | sg-0624d59607c3d632f   | 22         | TCP      | 205.254.168.94/32 | <a href="#">default</a> |

▼ Outbound rules

Filter rules

< 1 >

| Name | Security group rule ID | Port range | Protocol | Destination | Security groups         |
|------|------------------------|------------|----------|-------------|-------------------------|
| –    | sg-049a94bafb8ca6d20   | All        | All      | 0.0.0.0/0   | <a href="#">default</a> |

```

Last login: Sat Apr 26 10:54:35 2025 from 205.254.168.94
[ec2-user@ip-172-31-0-73 ~]$ sudo -i
[root@ip-172-31-0-73 ~]# yum install nginx -y
Last metadata expiration check: 19:40:32 ago on Sat Apr 26 10:55:18 2025.
Dependencies resolved.

```

| Package                  | Architecture | Version                 | Repository  | Size  |
|--------------------------|--------------|-------------------------|-------------|-------|
| Installing:<br>nginx     | x86_64       | 1:1.26.3-1.amzn2023.0.1 | amazonlinux | 33 k  |
| Installing dependencies: |              |                         |             |       |
| perf-tools-libs          | x86_64       | 2.9.1-1.amzn2023.0.3    | amazonlinux | 308 k |
| libwind                  | x86_64       | 1.4.0-5.amzn2023.0.2    | amazonlinux | 66 k  |
| nginx-core               | x86_64       | 1:1.26.3-1.amzn2023.0.1 | amazonlinux | 670 k |
| nginxfilesystem          | noarch       | 1:1.26.3-1.amzn2023.0.1 | amazonlinux | 9.6 k |
| nginx-mimetypes          | noarch       | 2.1.49-3.amzn2023.0.3   | amazonlinux | 21 k  |

Transaction Summary

```
=====
Install 6 Packages
```

Total download size: 1.1 M

Installed size: 3.6 M

```

Downloading Packages:
60 (2) - 3.1k

```

```
(1/6): libunwind-1.4.0-5.amzn2023.0.2.x86_64.rpm
(2/6): nginx-1.26.3-1.amzn2023.0.1.x86_64.rpm
(3/6): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm
(4/6): nginxfilesystem-1.26.3-1.amzn2023.0.1.noarch.rpm
(5/6): nginx-core-1.26.3-1.amzn2023.0.1.x86_64.rpm
(6/6): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm
```

|        |                                                   |          |        |       |
|--------|---------------------------------------------------|----------|--------|-------|
| (2/6): | nginx-1.26.3-1.amzn2023.0.1.x86_64.rpm            | 887 kB/s | 33 KB  | 00:00 |
| (3/6): | gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm   | 5.7 MB/s | 308 KB | 00:00 |
| (4/6): | nginx-filesystem-1.26.3-1.amzn2023.0.1.noarch.rpm | 441 kB/s | 9.6 KB | 00:00 |
| (5/6): | nginx-core-1.26.3-1.amzn2023.0.1.x86_64.rpm       | 20 MB/s  | 670 KB | 00:00 |
| (6/6): | nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm  | 1.2 MB/s | 21 KB  | 00:00 |

-----  
Total

Running transaction check

Transaction check succeeded.

Running transaction test  
Transaction test passed

Transaction test succeeded.  
Running transaction

Running Transaction  
Preparing

### Running scriptlet

## Installing

## Installing

## Installing

## Installing

Installing  
Installing

## Installing Running scriptlet

Verifying

Verifying

## Verifying

## Verifying

Verifying  
Verifying

### Verifying

Installed:

```
gperftools-libs-2
```

```
nginx-core-1:1.26
```

1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 26

Complete!

```
[root@ip-172-31-0-7 ~]#
```

```
-bash: !: event not
[root@ip-172-31-0-7
```

```

[Root@ip-172-31-0-7
updated adm/

```

```

account/  admin/
cache/

```

```
[root@ip-172-31-0-7
```

[README](#) [audit](#) [chro](#)

amazon btmp clou

```
[root@ip-172-31-0-7
```

```
ls: cannot access '':
[root@ip-172-31-0-7
```

```
[root@ip-172-31-0-7 ~]#
```

```

[1000@ip-172-31-0-7
cgi-bin html]

```

```
[root@ip-172-31-0-7
```

Hello from Server 1

```
[root@ip-172-31-0-7
```

Step 2: Go to ec2 dashboard and create an application load balancer which allows http traffic and configure the required load balancer settings.

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Mourya

EC2

Load balancers

Create Application Load Balancer

Application Load Balancers now support public IPv4 IP Address Management (IPAM)

You can get started with this feature by configuring IP pools in the Network mapping section.

## Create Application Load Balancer [Info](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

### ► How Application Load Balancers work

#### Basic configuration

##### Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

##### Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

###### ☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

###### ☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

##### Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

###### ☒ IPv4

Includes only IPv4 addresses.

###### ☐ Dualstack

Includes IPv4 and IPv6 addresses.

###### ☐ Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

EC2

Load balancers

Create Application Load Balancer

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#). For a new VPC, [create a VPC](#).

-

vpc-05c709098162d5641

IPv4 VPC CIDR: 172.31.0.0/16

IP pools - new [Info](#)

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-09f54f91a41232341

IPv4 subnet CIDR: 172.31.0.0/20

☒ ap-south-1b (aps1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0cd1ec5e21897c12d

IPv4 subnet CIDR: 172.31.0.0/20

☐ ap-south-1c (aps1-az2)

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

EC2

>

Load balancers

>

Create Application Load Balancer

Security groups

info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

default

sg-3d0e6e6d918fd6d4

VPC: vpc-03c709098162d5641

Listeners and routing

info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Remove

Protocol

HTTP

Port

80

1-65535

Default action

Info

Forward to

Select a target group

Create target group

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

Add listener

► Load balancer tags - optional

Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them. The 'Key' is required, but 'Value' is optional. For example, you can have Key = production-webserver, or Key = webserver, and Value = production.

Step 3: Create a target group and register both the ec2 instances in this target group.

EC2

>

Target groups

>

Create target group

Step 1

Specify group details

Step 2

Register targets

Specify group details

Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

Instances

- Supports load balancing to instances within a specific VPC.
- Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

IP addresses

- Supports load balancing to VPC and on-premises resources.
- Facilitates routing to multiple IP addresses and network interfaces on the same instance.
- Offers flexibility with microservice based architectures, simplifying inter-application communication.
- Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.

Lambda function

- Facilitates routing to a single Lambda function.
- Accessible to Application Load Balancers only.

Application Load Balancer

- Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
- Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name

website-hosting1g

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port



Amazon EC2

Search [Alt+Q]

Target groups > Create target group

**Protocol : Port**  
Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation.

HTTP 80 1-65535

**IP address type**  
Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4  
Each instance has a default network interface (eni) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6  
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eni). [Learn more](#)

**VPC**  
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-03c79988b1c2d5a41  
Private VPC CIDR: 10.0.0.0/16

**Protocol version**

☒ HTTP1  
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2  
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC specific features are not available.

☐ gRPC  
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

**Health checks**  
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

**Health check protocol**  
HTTP

**Health check path**  
Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/

Up to 1024 characters allowed.

Advanced health check settings

**Attributes**

☐ Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

Tags - optional  
Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel Next

EC2 > Target groups > Create target group

Step 1  
Specify group details

Step 2  
Register targets

**Register targets**  
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

**Available instances (2)**

Filter instances

| <input type="checkbox"/> | Instance ID         | Name                    | State   | Security groups | Zone        | Private IP-v4 address | Subnet ID                | La |
|--------------------------|---------------------|-------------------------|---------|-----------------|-------------|-----------------------|--------------------------|----|
| <input type="checkbox"/> | i-05505co0128cf8d24 | Multi-Website Hosting 1 | Running | default         | ap-south-1b | 172.31.0.73           | subnet-0cd1ec5e21897c12d | Ap |
| <input type="checkbox"/> | i-0b83da39386aa114a | Multi-Website Hosting 2 | Running | default         | ap-south-1b | 172.31.2.251          | subnet-0cd1ec5e21897c12d | Ap |

0 selected

**Ports for the selected instances**  
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

2 selections are now pending below. Include more or register targets when ready.

**Review targets**

**Targets (2)**

Filter targets Show only pending Remove all pending

| Instance ID         | Name                    | Port | State   | Security groups | Zone        | Private IP-v4 address | Subnet ID                | Launch time                       |
|---------------------|-------------------------|------|---------|-----------------|-------------|-----------------------|--------------------------|-----------------------------------|
| i-05505co0128cf8d24 | Multi-Website Hosting 1 | 80   | Running | default         | ap-south-1b | 172.31.0.73           | subnet-0cd1ec5e21897c12d | April 27, 2025, 11:57 (UTC+05:30) |
| i-0b83da39386aa114a | Multi-Website Hosting 2 | 80   | Running | default         | ap-south-1b | 172.31.2.251          | subnet-0cd1ec5e21897c12d | April 27, 2025, 11:57 (UTC+05:30) |

2 pending

Cancel Previous Create target group

Step 4: Attach the target group to the load balancer you just created.

EC2 > Load balancers > Create Application Load Balancer

Listener: H1 1P25U

Protocol: HTTP Port: 80

Default action: Forward to website-hostingTG (Target type: Instance, IPv4)

Listener tags - optional

Load balancer tags - optional

Optimize with service integrations - optional

Amazon CloudFront + AWS Web Application Firewall (WAF) - new

AWS Web Application Firewall (WAF)

AWS Global Accelerator

Step 5: Find your load balancer's DNS name and copy, paste it in to the browser.

Application Load Balancers now support public IPv4 IP Address Management (IPAM)

website-hostingLB

Details

Load balancer type: Application

Status: Provisioning

VPC: vpc-03c709098162d5641

Load balancer IP address type: IPv4

Scheme: Internet-facing

Hosted zone: ZP97RAF1XTN2K

Availability Zones: subnet-09f54f91a41232341 (ap-south-1a), subnet-0cd1ec5e21897c12d (ap-south-1b)

Load balancer ARN: arn:aws:elasticloadbalancing:ap-south-1:182563999454:loadbalancer/app/website-hostingLB/7d3bf2ba9ff77ed8

website-hostingLB-1593605813.ap-south-1.elb.amazonaws.com (A Record)

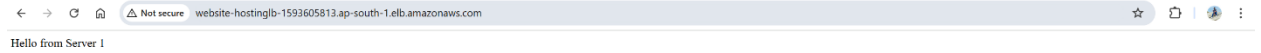
Listeners and rules

Listeners and rules (1)

Filter listeners

| Protocol:Port | Default action          | Rules  | ARN | Security policy | Default SSL/TLS certificate | mTLS           |
|---------------|-------------------------|--------|-----|-----------------|-----------------------------|----------------|
| HTTP:80       | Forward to target group | 1 rule | ARN | Not applicable  | Not applicable              | Not applicable |

Step 6: Your website should load.



**10. If one EC2 instance under a load balancer becomes unhealthy, what happens automatically? How can you monitor this?**

The load balancer will automatically stop sending traffic to the unhealthy ec2 instance

and will send users to only the healthy instance. This happens because load balancer will do regular health checks. You can monitor the health of ec2 instance by following the below steps.

1) Go to ec2 dashboard -> Load Balancer -> click on your load balancer.

The screenshot shows the AWS Management Console interface for the 'website-hostingLB' Load Balancer. The left sidebar contains navigation links for various AWS services. The main content area displays the 'Details' tab, which includes information about the Load Balancer type (Application), Scheme (Internet-facing), Status (Active), Hosted zone (ZP97RAFLXTNZK), VPC (vpc-03c709098162d5641), Availability Zones (subnet-09f54f91a41232341 and subnet-0cd1ec5e21897c12d), Load Balancer IP address type (IPv4), Date created (April 27, 2025), Load Balancer ARN, and DNS name info. Below the details, there are tabs for 'Listeners and rules', 'Network mapping', 'Resource map', 'Security', 'Monitoring', 'Integrations', 'Attributes', 'Capacity', and 'Tags'. The 'Listeners and rules' tab is selected, showing a single listener for HTTP:80 with a default action of 'Forward to target group'.

2) Click on target groups attached to your load balancer.

3) Under targets tab you can see healthy instances(in green), unhealthy(in red).

The screenshot shows the AWS Management Console interface for the 'website-hostingTG' Target Group. The left sidebar contains navigation links for various AWS services. The main content area displays the 'Details' tab, which includes information about the Target type (Instance), Protocol: Port (HTTP: 80), Protocol version (HTTP1), VPC (vpc-03c709098162d5641), IP address type (IPv4), Load balancer (website-hostingLB), and a summary of targets (2 Total targets, 2 Healthy, 0 Unhealthy, 0 Anomalous). Below the details, there are tabs for 'Targets', 'Monitoring', 'Health checks', 'Attributes', and 'Tags'. The 'Targets' tab is selected, showing a table of registered targets with columns for Instance ID, Name, Port, Zone, Health status, Health status details, Admini..., Overri..., and Launch... The table lists two healthy targets.

## Section 3: S3 Buckets and Static Websites (3 Questions)

### 11. How do you create an S3 bucket and upload a file using the AWS Management Console?

Step 1: Go to All services -> S3 or you can directly search s3 in the search bar on the top.



Step 2: Click on Create Bucket and configure required settings.

A screenshot of the AWS Management Console 'Create bucket' page. The page title is 'Create bucket' with an 'Info' link. Below the title, it says 'Buckets are containers for data stored in S3.' The page is divided into three main sections: 'General configuration', 'Object Ownership', and 'Block Public Access settings for this bucket'. In the 'General configuration' section, the 'AWS Region' is set to 'Asia Pacific (Mumbai) ap-south-1'. The 'Bucket type' is set to 'General purpose' (recommended for most use cases). The 'Bucket name' is 'mourya-samplebucket'. In the 'Object Ownership' section, 'ACLs disabled (recommended)' is selected. In the 'Block Public Access settings for this bucket' section, 'Block all public access' is selected. The page also includes a 'Choose bucket' button and a 'Format: s3://bucket/prefix' text.

**Block Public Access settings for this bucket**

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**  
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**  
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**  
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☐ **Block public access to buckets and objects granted through new public bucket or access point policies**  
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**  
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

**Bucket Versioning**

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

**Bucket Versioning**

☒ Disable  
☐ Enable

**Tags - optional (0)**

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

No tags associated with this bucket.

[Add tag](#)

Step 3: Click on your bucket name -> Upload -> Add files.

**Upload**

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

**Files and folders (0)**

All files and folders in this table will be uploaded.

| Name                | Folder | Type | Size |
|---------------------|--------|------|------|
| No files or folders |        |      |      |

You have not chosen any files or folders to upload.

**Destination**

**Destination**  
[s3://sample-bucket-mourya](#)

**Destination details**  
Bucket settings that impact new objects stored in the specified destination.

**Permissions**  
Grant public access and access to other AWS accounts.

**Properties**  
Specify storage class, encryption settings, tags, and more.

[Cancel](#) [Upload](#)

Step 4: Add a file from your local machine or select the source from which you can able to add the files.

- (List the important settings like region, permissions.)

Storage class: Leave default unless you want cheaper storage (like Glacier).

Region: Choose the region closest to your users (ex: ap-south-1, us-east-1).

Permissions: By default files are private, you can make them public if needed by disabling block public access.

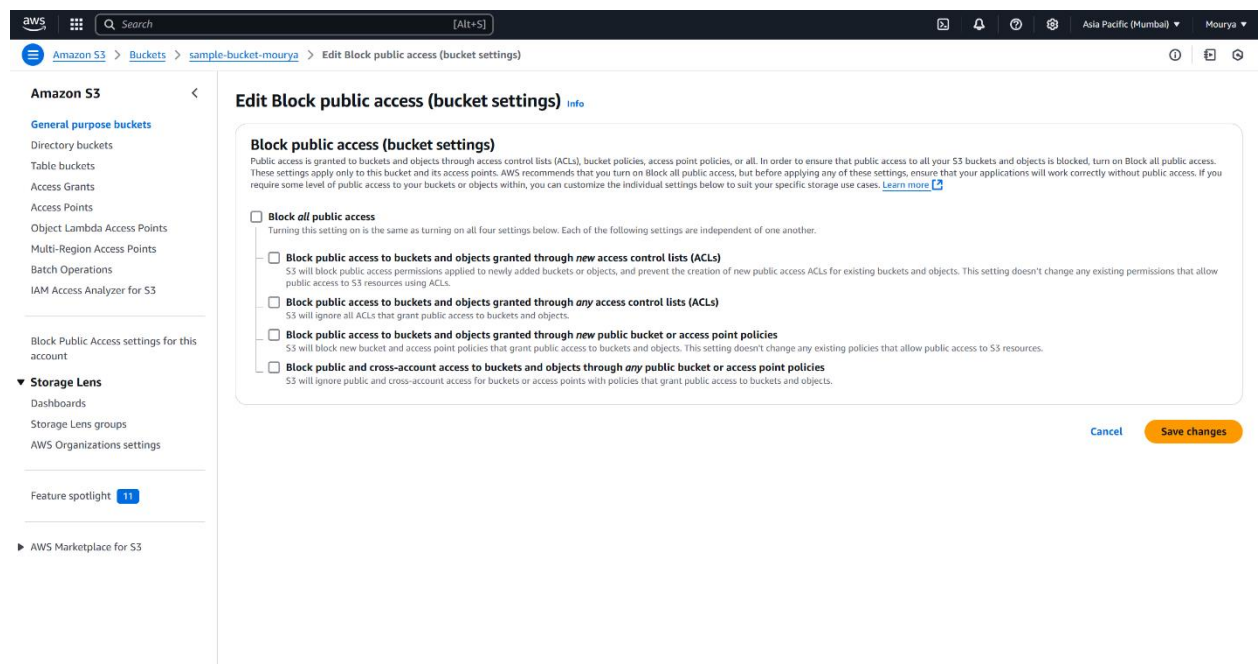
Versioning: Turn ON if you want to keep versions of your files (Useful for backups).

Encryption: Turn ON if you want to encrypt your files while uploading.

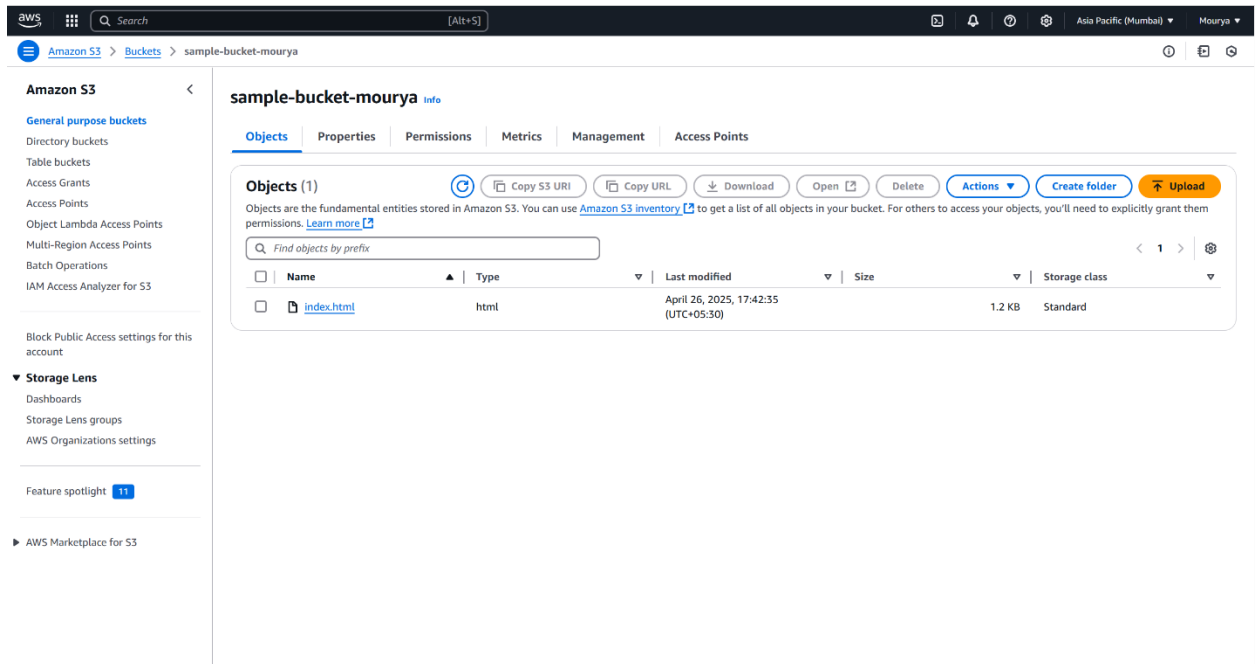
## 12. How would you host a static website using an S3 bucket?

- (Mention website hosting option, bucket policy for public access.)

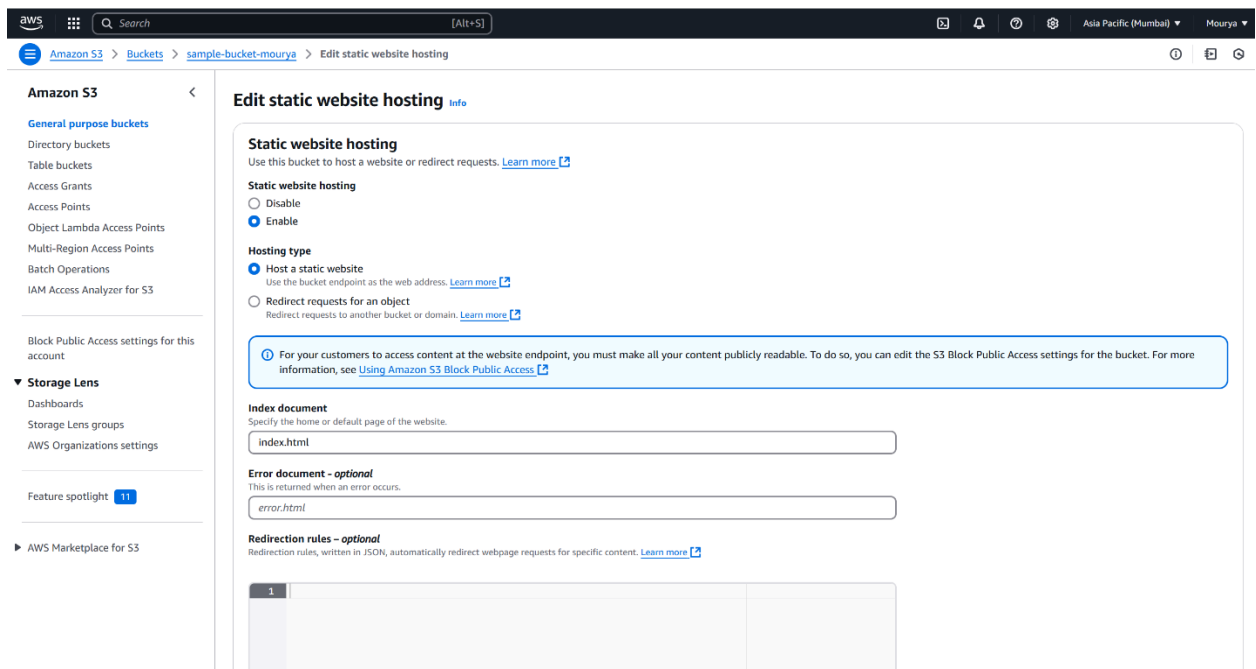
Step 1: Create an s3 bucket and uncheck 'block all public access', because websites need to be public.



Step 2: Open your bucket -> Upload -> Add files, add your required website files (ex: index.html, styles.css e.t.c).

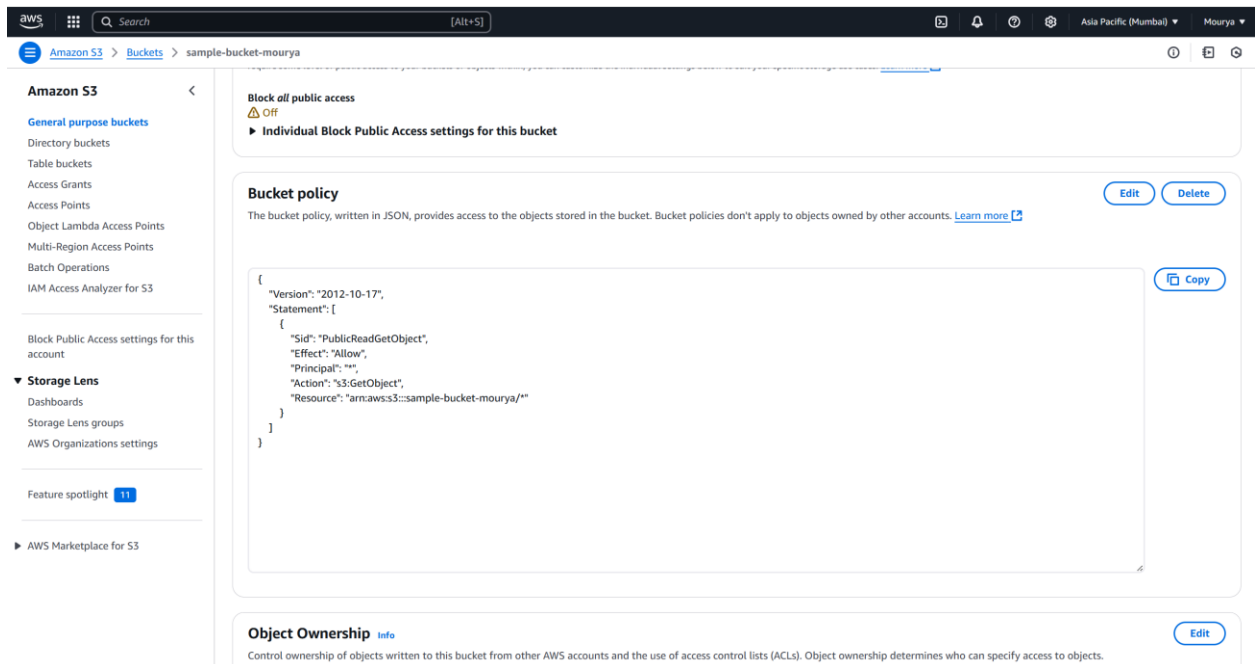


Step 3: Go to Properties -> Static Website Hosting -> Edit -> Enable it and fill the required details.

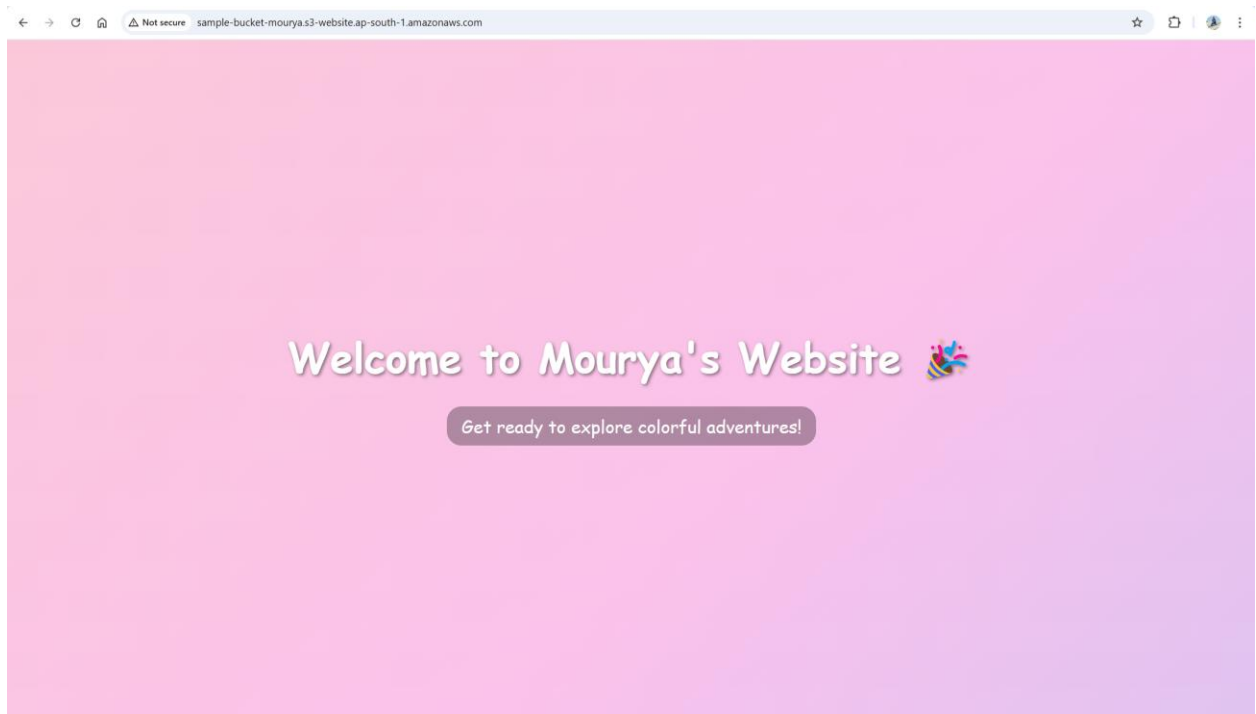


Step 4: Go to Permissions -> Bucket Policy -> Edit -> write the policy for public access.





Step 5: Go back to Permissions -> under Static website hosting you will find the website url, open it the browser and you can able to see your static website.



### 13. What are the major differences between S3 Standard Storage Class and S3 Glacier Storage Class?

S3 Standard is a storage class where you can store the files (objects) that you might

need anytime quickly. It is designed for frequent access, good for websites, apps, daily reports. It is little expensive compared to glacier.

S3 Glacier is a long term and rare access storage class. Its like putting your stuff deep inside a storage room. If you want to access the files from it, you need to wait for a while. It is cheaper compared to Standard storage class. It is good for the files rarely you need like old backup file, legal documents e.t.c.

## Section 4: Shell Scripting - Nginx Setup (2 Questions)

14. Write a shell script that installs **nginx** on an Ubuntu EC2 instance and sets up two websites (**site1.com** and **site2.com**) using different server blocks (virtual hosts).

Create an Ec2 Instance and connect to it from your local machine through git bash.  
Create a file and write the required code for installing nginx and set up two websites.

```

root@ip:172.31.1-244~
#1/bin/bash

# Update the package list
sudo yum update -y

# Install nginx
sudo yum install nginx -y

# Start nginx and enable it to run
sudo systemctl start nginx
sudo systemctl enable nginx

# Multiple Website Hosting

mkdir /usr/share/nginx/site1
mkdir /usr/share/nginx/site2

echo "<html><body><h1> Hello from Site1 </h1></body></html>" | sudo tee /usr/share/nginx/site1/index.html
echo "<html><body><h1> Hello from Site2 </h1></body></html>" | sudo tee /usr/share/nginx/site2/index.html

echo "server {
    listen 80;
    server_name site1.mourya.com;
    root /usr/share/nginx/site1;
}" | sudo tee /etc/nginx/conf.d/site1.conf

echo "server {
    listen 80;
    server_name site2.mourya.com;
    root /usr/share/nginx/site2;
}" | sudo tee /etc/nginx/conf.d/site2.conf

# Checking for Syntax errors
nginx -t

# Reloading Nginx
systemctl reload nginx

```

Now in your local machine go to C:\Windows\System32\drivers\etc\hosts, open the hosts file as administrator and then aim your website url to your server ipv4.

## 15. What changes would you need to make to ensure that nginx starts automatically after server reboot?

- (Mention enabling services.)

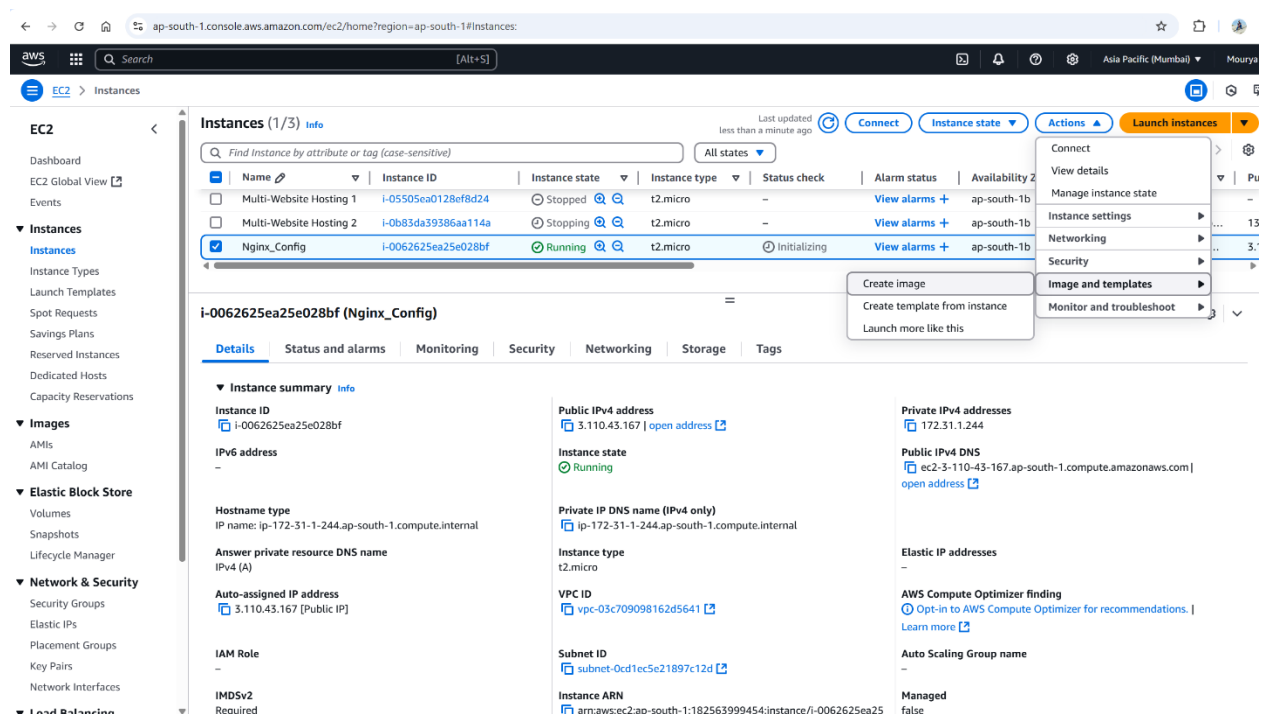
-> systemctl start nginx (it will start nginx, if it's not running after the server reboot)

-> systemctl enable nginx (it will start nginx automatically whenever the server gets reboot).

## Section 5: AWS AMIs and Volume Snapshots (2 Questions)

### 16. Explain the steps to create an Amazon Machine Image (AMI) from an existing EC2 instance.

1) Go to ec2 instance -> select & start your instance -> Click on actions -> image and templates -> create image



2) Fill image details and click on create image.

**Create image** [Info](#)

An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.

**Instance ID**  
[i-00c2625ca25e028bf](#) (Nginx\_Config)

**Image name**

Maximum 127 characters. Can't be modified after creation.

**Image description - optional**

Maximum 255 characters

☒ **Reboot instance**  
 When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

**Instance volumes**

| Storage type | Device     | Snapshot                      | Size | Volume type                   | IOPS | Throughput | Delete on termination                      | Encrypted                       |
|--------------|------------|-------------------------------|------|-------------------------------|------|------------|--------------------------------------------|---------------------------------|
| EBS          | /dev/sd... | Create new snapshot from v... | 8    | EBS General Purpose SSD - ... | 3000 |            | <input checked="" type="checkbox"/> Enable | <input type="checkbox"/> Enable |

[Add volume](#)

During the image creation process, Amazon EC2 creates a snapshot of each of the above volumes.

**Tags - optional**  
 A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

☒ **Tag image and snapshots together**  
 Tag the image and the snapshots with the same tag.

☐ **Tag image and snapshots separately**  
 Tag the image and the snapshots with different tags.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Create image](#)

3) Now go to ec2 -> AMIs, you can see the image you have create. Wait for a while until it changes to available and then you can able to use it.

**Amazon Machine Images (AMIs) (1)** [Info](#)

[Owned by me](#) [Find AMI by attribute or tag](#)

|                          | Name               | AMI name | AMI ID                | Source                          | Owner        | Visibility | Status    |
|--------------------------|--------------------|----------|-----------------------|---------------------------------|--------------|------------|-----------|
| <input type="checkbox"/> | nginx-config-image |          | ami-009c5e09e43dc7513 | 182563999454/nginx-config-image | 182563999454 | Private    | Available |

Select an AMI

## 17. What are EBS volume snapshots? How can you automate regular snapshots for backup purposes?

EBS (Elastic Block Storage) volume is like a hard disk attached to your instance, it is used to store your data

EBS Snapshot is like taking a snap of our EBS volume, it is used to store backup copy of the ec2 server

Automate regular snapshots:

1) Go to ec2 -> life cycle manager -> create new ebs snapshot policy.

The screenshot shows the AWS Management Console interface for the Amazon Data Lifecycle Manager (DLM) console. The left-hand navigation pane lists various AWS services, with 'Elastic Block Store' and 'Lifecycle Manager' highlighted. The main content area displays the 'Amazon Data Lifecycle Manager' header and a sub-header 'Automate the creation, retention, copy and deletion of snapshots and AMIs'. A 'Create new lifecycle policy' dialog box is open, allowing the user to choose between 'Custom policy' (selected) and 'Default policy'. The 'Policy type' dropdown menu is set to 'EBS snapshot policy'. Below the main content, there are sections for 'Benefits and features' and 'Pricing'. The 'Benefits and features' section includes 'Automated snapshot and AMI creation', 'Fast snapshot restore integration', 'Built-in cross-Region copy', and 'Automated cross-account snapshot copy'. The 'Pricing' section states that there is no charge for using Amazon Data Lifecycle Manager. A 'Getting started' section with links to AWS Blog articles is also visible.

2) Configure the policy settings, review and create the policy.

Step 1: Specify settings

**Specify settings**

**Target resources** [info](#)  
Specify the resources that are to be targeted by this policy.

**Target resource types**  
Select the type of resources that are to be targeted.

☒ Volume  
☐ Instance

**Target resource tags**  
All resources of the selected types that have at least one of these tags will be targeted by the policy.

[Add](#)

[policy](#) [x](#)  
policy  
44 tags remaining of 45.

**Description**  
Policy description  
Nginx\_Config\_Policy

**IAM role** [info](#)  
This policy must be associated with an IAM role that has the appropriate permissions. If you choose to create a new role, you must grant relevant role permissions and set up trust relationships correctly. If you are unsure of what role to use, choose Default role.

☒ Default role  
[If the default role already exists, Amazon Data Lifecycle Manager will use that role. If it does not exist yet, it will be automatically created with all the required permissions.](#)

[View default role permissions](#)  
☐ Choose another role

**Tags - optional** [info](#)  
Assign custom tags to the policy to help you identify, organize, and secure your lifecycle policies. Each tag consists of a key and an optional value. No tags associated with the resource.

[Add tag](#)  
You can add 25 more tags.

**Policy status**  
Specify whether to enable the policy immediately after creation or modification. If you do not enable the policy now, then it will not begin creating snapshots or AMIs until you manually set its activation status to enabled.

☒ Enabled  
☐ Not enabled

Step 2: Configure schedule 1 - Schedule 1

**Configure schedule 1 - Schedule 1**

Schedules define how often the policy runs and the specific actions that are to be performed. The policy must have at least one schedule.

[Additional schedules must have the same retention type as Schedule 1, but they can have their own retention count or age. Snapshot archiving can be enabled for one schedule only.](#)

[Remove schedule](#) [Add another schedule](#)  
You can add 3 more schedules to this policy.

**Schedule details** [info](#)

**Schedule name**  
Schedule 1

**Frequency**  
Daily

**Every**  
12 hours

**Starting at**  
09:00 UTC

**Retention type**  
Count

**Keep**  
14 snapshots in standard tier

**Advanced settings - optional**

- Tagging** [info](#)  
Specify the tags that are to be applied to snapshots created by this schedule. These tags are not applied to cross-Region copies created by the schedule.
- Snapshot archiving** [info](#)  
Enable snapshot archiving to automatically move snapshots created by this schedule from the standard storage tier to the archive storage tier.
- Fast snapshot restore** [info](#)  
Enable fast snapshot restore to ensure that volumes created from snapshots created by this schedule instantly deliver all of their provisioned performance.
- Cross-Region copy** [info](#)  
Enable cross-Region copy to copy snapshots created by this schedule to up to three additional Regions.
- Cross-account sharing** [info](#)  
Enable cross-account sharing to share the snapshots created by this schedule with other AWS accounts.

[Cancel](#) [Previous](#) [Review policy](#)

3) AWS will now automatically take the snapshots by deleting the old ones according to your policy.



## Final Submission Task:

- Upload all your answers in a single folder.
- Include your shell script separately.
- Push your work to GitHub.

Raise a **Pull Request** to the following repository:



**Repository URL:** <https://github.com/Kloudvidhya/assignments.git>

---

Good Luck! 🍀

#DevOps #AWS #GitHub #Kloudvidhya